

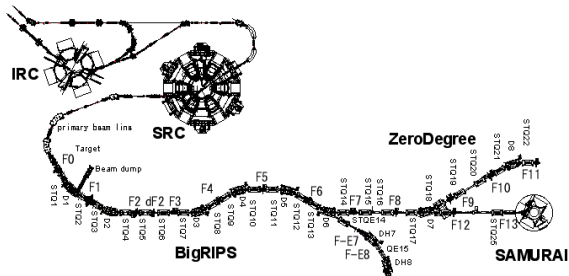
Why SAMURAI73 Needs Two Polarimeters

Full tensor reconstruction and locating p_{yy} loss

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3 September 2026

IVR needs the tensor state delivered to the target



First IVR run

Start with vertical p_{yy} ;
longitudinal p_{zz} is harder
to prepare.

After SRC

What polarization leaves
the accelerator chain?

After BigRIPS

What polarization reaches
SAMURAI73?

Two stations are needed to determine all degrees of freedom of the spin-density matrix.

Spin-1: 8 general degrees, 5 tensor, 1 prepared

For $w_a \geq 0$ and $\sum_a w_a = 1$,

$$\rho = \sum_a w_a |\psi_a\rangle \langle \psi_a|.$$

A normalized 3×3 Hermitian ρ has

$$8 = 3_{\text{vector}} + 5_{\text{tensor}}$$

independent real parameters.

An axial tensor prepared about $\hat{n}$ is

$$\mathbf{P}_0^{(2)}(q, \hat{n}) = \frac{q}{2} (3\hat{n}\hat{n}^T - I).$$

$$\hat{n} = \hat{y} : \quad \mathbf{P}_0^{(2)} = \frac{p_{yy}}{2} \text{diag}(-1, 2, -1),$$

$$\hat{n} = \hat{z} : \quad \mathbf{P}_0^{(2)} = \frac{p_{zz}}{2} \text{diag}(-1, -1, 2).$$

Tensor only: 5 degrees. Fixed preparation axis: one amplitude q .

Thomas–BMT transport is a rotation

For a magnetic lattice ($\mathbf{E} = 0$), each trajectory obeys the Thomas–BMT equation

$$\frac{d\mathbf{p}}{dt} = \boldsymbol{\Omega}_B \times \mathbf{p}, \quad \boldsymbol{\Omega}_B = -\frac{q}{m} \left[\left(G + \frac{1}{\gamma} \right) \mathbf{B} - \frac{G\gamma}{\gamma+1} (\boldsymbol{\beta} \cdot \mathbf{B}) \boldsymbol{\beta} \right], \quad G = \frac{g-2}{2}.$$

The cross product generates a rotation

$$K_B \mathbf{v} \equiv \boldsymbol{\Omega}_B \times \mathbf{v}, \quad K_B^T = -K_B,$$

$$\dot{R}_s = K_B R_s,$$

$$R_s(t_f, t_0) = \mathcal{T} \exp \left[\int_{t_0}^{t_f} K_B dt \right] \in SO(3).$$

Both spin moments follow the same R_s

$$\mathbf{p}_f = R_s \mathbf{p}_0, \quad \mathbf{P}_f^{(2)} = R_s \mathbf{P}_0^{(2)} R_s^T.$$

The downstream momentum supplies a passive change of axes:

$$R_{\text{rel}} = R_o^T R_s, \quad \mathbf{p}_{b,f} = R_{\text{rel}} \mathbf{p}_0,$$

$$\mathbf{P}_{b,f}^{(2)} = R_{\text{rel}} \mathbf{P}_0^{(2)} R_{\text{rel}}^T.$$

BMT precession applies one $SO(3)$ rotation to both vector and tensor moments.

BMT spread mixes tensors and can reveal a vector

A zero entrance mean does not require zero vector polarization in every phase-space slice:

$$\bar{\mathbf{p}}_0 = \int d\xi f(\xi) \mathbf{p}_0(\xi) = 0, \quad \overline{\mathbf{P}^{(2)}}_0 \neq 0, \quad \mathbf{p}_0(\xi) \neq 0.$$

Entrance: two slice vectors cancel

$$\mathbf{p}_0^{(+)} = p \hat{\mathbf{x}}, \quad \mathbf{p}_0^{(-)} = -p \hat{\mathbf{x}},$$

These give $\bar{\mathbf{p}}_0 = 0$. Their trajectory maps are $R_+ = R_z(+\theta)$ and $R_- = R_z(-\theta)$.

Exit: the rotated vectors add

$$\begin{aligned} \bar{\mathbf{p}}_f &= \frac{1}{2} \left(R_+ \mathbf{p}_0^{(+)} + R_- \mathbf{p}_0^{(-)} \right) \\ &= p \sin \theta \hat{\mathbf{y}} \neq 0. \end{aligned}$$

The vector already existed slice by slice.

The same phase-space-dependent rotations mix the five tensor components:

$$\overline{\mathbf{P}^{(2)}}_f = \int d\xi f(\xi) R_\xi \mathbf{P}_0^{(2)}(\xi) R_\xi^T,$$

The beam-average tensor can become biaxial. Local rank conversion requires higher-order spin coupling.

Correlated BMT rotations can expose initially cancelling slice vectors.

Rank mixing opens all 8 state parameters

Local tensor–vector conversion requires spin dynamics beyond the linear BMT term.
For example,

$$H_{\text{spin}} = -\mu \mathbf{S} \cdot \mathbf{B} - \beta_T (\mathbf{S} \cdot \mathbf{B})^2 + \dots, \quad \bar{\rho}_f = \int d\xi f(\xi) \mathcal{E}_\xi[\rho_0].$$

Quadratic spin interaction

Tensor magnetic/electric polarizability or other quadratic spin couplings can generate

$$\mathbf{p}_0 = 0, \mathbf{P}_0^{(2)} \neq 0 \quad \longrightarrow \quad \mathbf{p}_f \neq 0.$$

Spin-dependent loss or material interaction

Spin-dependent loss, material interactions, collimation, and acceptance can change both ranks and the density-matrix eigenvalues.

Once rank mixing is allowed, the fit must allow the full $8 = 3_{\text{vector}} + 5_{\text{tensor}}$ parameters.

The cross section shows the two blind directions

$$\begin{aligned}\frac{\sigma(\theta, \phi)}{\sigma_0(\theta)} = & 1 + \frac{3}{2} (p_x \sin \phi + p_y \cos \phi) A_y(\theta) \\ & + \frac{2}{3} (p_{xz} \cos \phi - p_{yz} \sin \phi) A_{xz}(\theta) \\ & + \frac{1}{6} [(p_{xx} - p_{yy}) \cos 2\phi - 2p_{xy} \sin 2\phi] [A_{xx}(\theta) - A_{yy}(\theta)] \\ & + \frac{1}{2} p_{zz} A_{zz}(\theta).\end{aligned}$$

p_z is absent

The d - p reaction has no longitudinal-vector term in this geometry.

One calibrated multi-angle station measures 6/8 density-matrix parameters, or 4/5 tensor parameters.

p_{xy} vanishes at the four arms

At L/R/U/D, $\phi = 0^\circ, 90^\circ, 180^\circ, 270^\circ$, so $\sin 2\phi = 0$.

BigRIPS makes both blind directions visible

Before BigRIPS: unprimed.

After BigRIPS: primed.

$$R_{\text{rel}} = R_y(\delta), \quad \delta = 27.513^\circ, \quad c = \cos \delta = 0.886906, \quad s = \sin \delta = 0.461951.$$

Vector blind direction

$$\begin{aligned} p'_x &= c p_x + s p_z, \\ p'_z &= -s p_x + c p_z. \end{aligned}$$

Hence an upstream p_z gives

$$\boxed{p'_x = 0.461951 p_z},$$

which is measured at station 2.

Two stations: tensor 5/5; full density matrix 8/8.

Tensor blind direction

$$\begin{aligned} p'_{xy} &= c p_{xy} + s p_{yz}, \\ p'_{yz} &= -s p_{xy} + c p_{yz}. \end{aligned}$$

Hence an upstream p_{xy} gives

$$\boxed{p'_{yz} = -0.461951 p_{xy}},$$

which is measured at station 2.

Two stations localize changes in the spin state

The BMT reference at station 2 must retain the slice distribution:

$$\bar{\rho}_{2,\text{BMT}} = \int d\xi f_2(\xi) U_\xi \rho_1(\xi) U_\xi^\dagger.$$

The measured average $\bar{\rho}_1$ alone omits correlations between spin and phase space.

station comparison	location of the change
$\bar{\rho}_1 \not\simeq \rho_{\text{prep}}$	The source, acceleration chain, or SRC changed the state.
$\bar{\rho}_1 \simeq \rho_{\text{prep}}$ and $\bar{\rho}_2 \not\simeq \bar{\rho}_{2,\text{BMT}}$	The state changed between the two stations.

Mechanism: The BMT integral includes field spread and hidden slice vectors. Any remaining mismatch tests higher-order or spin-selective dynamics.

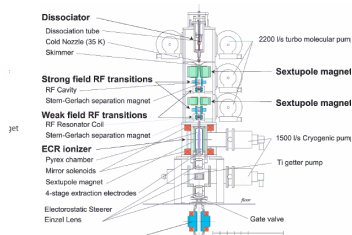
After-SRC installation: contact and procedure

1. Who should we contact at RIKEN?
2. What approval and installation procedure should we follow?

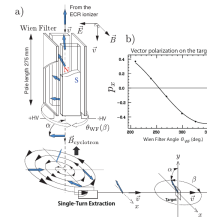
Backup

Source, accelerator, beam-line, and detector details

Backup: Wien filter sets the PIS spin axis



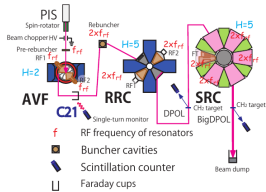
RIKEN polarized ion source



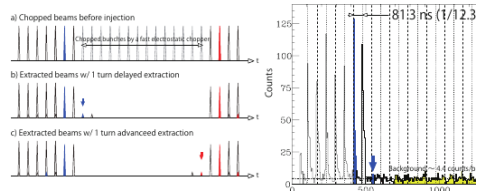
Wien-filter spin-orientation control

- ▶ The two-stage PIS produces vector and tensor modes.
- ▶ After the ECR ionizer the axis is vertical; crossed $\mathbf{E}$ and $\mathbf{B}$ tilt the spin without changing the orbit.
- ▶ Rotating the Wien filter sets the azimuth; vector polarization calibrates the angles.

Backup: single-turn extraction preserves spin phase



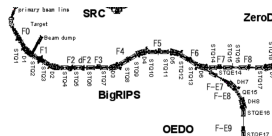
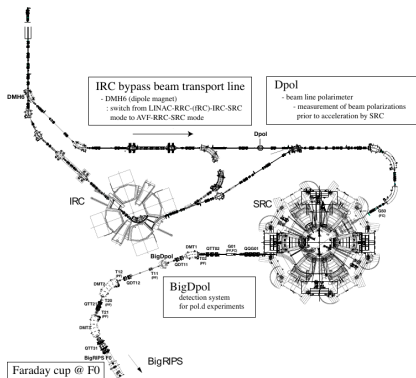
PIS → AVF → RRC → SRC



Timing signature of wrong-turn extraction

- ▶ The cyclotron field changes the spin direction every turn: $\Delta\beta = 360^\circ\gamma(g - 1)$, with $g = 0.8477$ in the paper's convention.
- ▶ Mixing N with $N \pm 1$ turns superposes spin phases and lowers the polarization.
- ▶ AVF, RRC, and SRC kept the same turn: 0.07% mixing, > 99% purity, stable for 2–3 days.

Backup: magnets between the two stations



- ▶ Station 1: BigDpol/T11, between DMT1 and DMT2.
- ▶ DMT2 and DMT3: two identical 50° dipoles.
- ▶ BigRIPS $F0 \rightarrow F7$: six 30° dipoles, signs $(-, -, -, +, +, -)$, net bend -60° .
- ▶ $F7 \rightarrow F13$: quadrupoles, but no incident-beam dipole before the target.

Backup: central ray gives $\delta = 27.513^\circ$ at 190 MeV/u

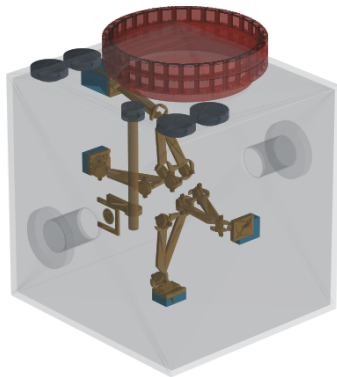
DMT2 + DMT3	-100°	For a 190 MeV/u deuteron, $\gamma = 1.20260044,$ $G_d = -0.14298727,$
BigRIPS $F0 \rightarrow F7$	-60°	
$F7 \rightarrow F13$	0°	
total orbit bend θ	-160°	$\delta = G_d \gamma \theta = +27.513^\circ,$ $R_{\text{rel}} = R_y(\delta).$

The central rotation exposes the one-station blind directions:

$$p'_x = 0.461951 p_z + \cdots, \quad p'_{yz} = -0.461951 p_{xy} + \cdots.$$

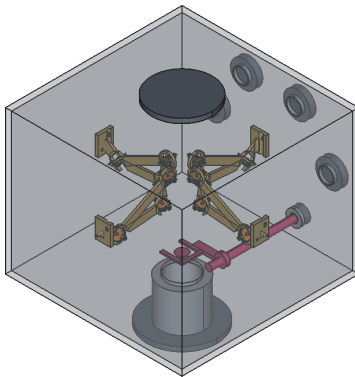
This is the central-ray horizontal-bend baseline; the real analysis uses $R_{\text{rel}}(\xi)$ from field maps and measured beam phase space.

Backup: proposed polarimeter after SRC



Compact in-vacuum polarimeter proposed just downstream of SRC.

Backup: proposed polarimeter before SAMURAI



Compact in-vacuum polarimeter proposed upstream of SAMURAI.

Backup: expected p_{yy} precision and rates

Detector geometry

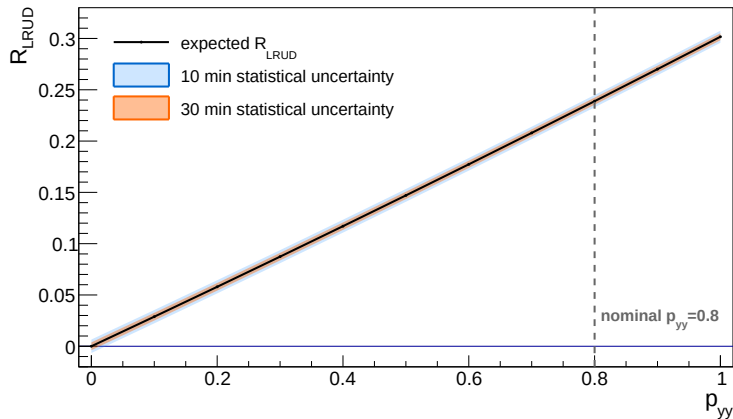
channel	θ_{lab}	radius
deuteron	20.9°	140 mm
proton, small angle	11.2°	190 mm
proton, large angle	53.4°	205 mm

Rate assumptions

beam	190 MeV/u, 10^7 d/s
nominal polarization	$p_{yy} = 0.8$
CH ₂ for rate model	1000 mg/cm ²
selected LR/UD rate	36.95 s^{-1}

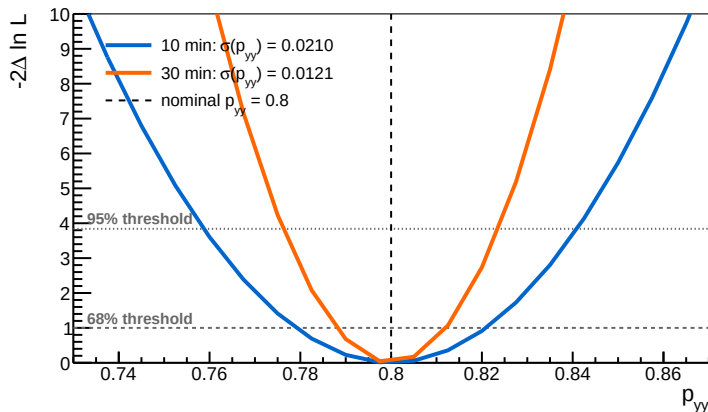
time	$N_L + N_R$	$N_U + N_D$	total	p_{yy} statistical precision
10 min	13,734	8,438	22,172	$\sigma = 0.0210$; 68% CI [0.779, 0.821]
30 min	41,203	25,314	66,517	$\sigma = 0.0121$; 68% CI [0.788, 0.812]

Backup: LR/UD count ratio versus p_{yy}



At $p_{yy} = 0.8$: $R_{LRUD} = 0.2389$; $\sigma_R = 0.0060$ (10 min) and 0.0035 (30 min).

Backup: p_{yy} precision at 10 and 30 min



At $p_{yy} = 0.8$: $\sigma = 0.0210$ at 10 min and 0.0121 at 30 min (42% smaller).